

What fails, and why: negative results on the Umans–Wang (α, β) -divisor conjecture

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Abstract

This companion note documents the approaches we tried against the Umans–Wang (α, β) -divisor conjecture that *failed*, and—more usefully—why each failed. The central empirical lesson is that every elementary “barrier” argument we produced was, on exact re-examination, an artifact: a generic/average magnitude estimate mistaken for a lower bound, a degenerate solution, or a solver timeout misread as infeasibility. We catalogue these so that future work does not repeat them, and we map the construction families that do and do not help, localizing the one mechanism (algebraic factoring of differences) that gives genuine but only constant-factor benefit in $\mathbb{Z}$.

1 Scope

The main note [1] proves $2\beta + \alpha \geq 1$ and reduces the conjecture at fixed α to a lattice-clustering statement equivalent to the conjecture. Here we record the negative space around that result: the arguments that looked like progress and were not. All claims below were checked by exact computation; the recurring theme is that exact checks repeatedly overturned plausible heuristics.

2 The recurring artifact: generic bounds are not lower bounds

Every “barrier” (i.e. would-be refutation of the conjecture) we found reduced to the same error, in five guises.

Dead end 1 (Vertex-log / incident-log forcing). One argues: a coordinate s_i constrained modulo a product of primes of total $\log L$ “must” have magnitude $\approx e^L$, and since some vertex accumulates $L \gg n^\alpha$, the budget is exceeded. **Why it fails:** this is the Minkowski *average*, not a lower bound. Residue sharing—choosing neighbours’ values so the forced residue is small—defeats it. Exact search finds solutions below the average in every tested case.

Dead end 2 (Row-product forcing in the asymmetric $(S - T)$ model). The two-sided analogue: s_i is “forced” modulo the product of its row’s primes $\approx e^{n/2K}$. **Why it fails:** identical to the above. With two free sets, choosing the t_j pulls s_i strictly below the row product; exact search confirms the gap grows with K .

Dead end 3 (Degenerate covers). Small “solutions” returned by unconstrained search often place $s_i = t_j$, so the difference is 0 and “covers” a prime trivially and invalidly. **Why it fails:** a zero

difference is not a cover; forbidding $s_i = t_j$ (and requiring distinctness *within* each set only) removes these. Some genuine solutions survive, but the degenerate ones must be filtered first.

Dead end 4 (Timeout as infeasibility). An integer program for the minimum K or minimum magnitude returns “not solved” (a solver time-out); this is read as “infeasible,” inflating the apparent minimum and manufacturing a barrier. **Why it fails:** time-out is not infeasibility. Distinguishing **Optimal/Infeasible/Timeout** explicitly, and cross-checking small cases against brute force, dissolves the effect; at the only cleanly solved sizes the counting floor is in fact reachable.

Dead end 5 (Wrong magnitude regime). Small- n experiments (magnitude $\leq 2n$, or $n^{O(1)}$) are used to estimate the scaling exponent. **Why it fails:** the conjecture’s difficulty lives in the *bundled* regime, where each difference is a product of $\Theta(n^\alpha / \log n)$ primes $> n/2$ and magnitude is e^{n^α} . A difference $\leq 2n$ carries at most one prime $> n/2$, so no polynomial-magnitude experiment ever enters the regime that matters; observed exponents there are about the wrong quantity.

Observation 1. *The uniform moral: in this problem, no counting or forcing argument yields a valid magnitude lower bound, because the covering freedom (residue sharing across a coprime system) always beats the generic estimate. The only genuine obstruction is the lattice-clustering statement of [1], which is equivalent to the conjecture itself.* Any future “barrier” claim derived from a counting or forcing bound should be presumed an artifact until checked exactly.

3 Construction families: what helps and what does not

On the constructive (pro-conjecture) side we tested many explicit families, measuring how many primes of $(n/2, n]$ a typical difference bundles relative to a random set.

- **Perfect squares: the one $\mathbb{Z}$ win.** With $S = \{a_i^2\}$, differences factor as $a_i^2 - a_j^2 = (a_i - a_j)(a_i + a_j)$, giving two independent large factors and $\approx 1.4\text{--}2\times$ the bundling of a random set. This is a genuine mechanism but only a *constant* factor; it does not change the exponent.
- **Higher powers: no.** For $a^k - b^k$ with $k \geq 3$, staying within the magnitude budget forces the roots below $B^{1/k}$, collapsing the extra factors; bundling drops back to generic. The benefit peaks sharply at $k = 2$.
- **Factorials: structurally useless.** A prime $p > n/2$ divides $m!$ only if $m \geq p$, but $m! \leq B$ forces m small; factorials are smooth in *small* primes only, the wrong range.
- **Near-squares ($a \cdot b$, $a \approx b$): no.** Bundling is generic. This sharpens the squares result: the benefit comes from the *exact algebraic identity* $a^2 - b^2 = (a - b)(a + b)$, not merely from two roughly equal factors.
- **Fractal / digit-restricted / Beatty / Fibonacci: no.** These impose *additive* structure (small difference sets), which is the sum-product antagonist of the multiplicative richness covering needs. The golden ratio, despite appearing in the unit-separation lemma of [3], does not transfer: its role there is archimedean separation, unrelated to integer divisibility.
- **Naive $G \cdot P$ in $\mathbb{Z}$: no better than random.** A sparse geometric G is the known-inadequate Balog–Wooley example; a dense multiplicative G (smooth numbers) matches random, because in the accessible regime random already covers. The genuine $G \cdot P$ power requires the *unit lattice* of a number field, which has no faithful $\mathbb{Z}$ analogue.

Observation 2. *Every family that helps does so through one mechanism: making differences factor into several large independent pieces. In $\mathbb{Z}$ this tops out at two factors (squares); reaching $\Theta(n^\alpha/\log n)$ factors per difference is inherently multi-dimensional and lands in the number-field construction of [3]. The construction search is, in this sense, saturated: the ceiling in $\mathbb{Z}$ is understood, and the only route past it is the one identified in the main note.*

4 Computational limits

Exact methods (bitmask minimum- K search; brute-force covering; small integer programs) reach roughly $n \leq 34$ for minimum- K and $K \leq 3$ –8 for the congruence systems, before the exponential wall. The exact minimum- K sequence does show the gap to the counting floor staying in $\{0, 1\}$ up to $n = 34$ —mildly suggestive that the floor is reachable—but this is in the no-bundling regime and therefore not evidence about the conjecture’s true (bundled) regime. No computation available to us reaches that regime, which requires super-polynomial magnitudes; this is not a limitation of implementation but of the problem’s location.

5 Summary

The negative results converge on a single picture consistent with the main note: the elementary obstructions are mirages (five documented forms of one error), the $\mathbb{Z}$ -native constructions saturate at a constant-factor gain from the difference-of-squares identity, and the true difficulty is the number-field clustering question. Recording these dead ends is the practical contribution here: they are the specific paths a future attack can skip.

References

- [1] E. Schultz. *A magnitude lower bound and a lattice-clustering reformulation of the Umans–Wang (α, β) -divisor conjecture*. Companion paper, 2026.
- [2] C. Umans and S. Wang. *A number-theoretic conjecture implying faster algorithms for polynomial factorization and integer factorization*. arXiv:2511.10851, 2025.
- [3] T. F. Bloom, W. Sawin, C. Schildkraut, and D. Zhelezov. *The sum-product conjecture is false for real numbers*. arXiv:2605.28781, 2026.